Amending the Unamendable: The Governed-Amendment Problem for Deep-Time Interstellar AI Probes

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Working draft

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Working draft. Part of a series on a slow, self-replicating interstellar AI probe. Each earlier paper — the vehicle, the payload, the bootstrapping closure paper, and the governance paper — reaches this problem and brackets it as the deepest piece of unfinished business. This paper does not solve it. It maps why it is so hard, examines the candidate solutions and shows where each falls short, and argues that conservative immutability remains the defensible default — not because the problem is closed, but because the alternative is worse.

Abstract

The payload paper resolves the central tension of a deep-time AI mind — how to let capability grow without letting purpose drift — by placing the probe's founding goals and values in an **immutable core** that every replica must reproduce exactly, while a large mutable cognitive layer is left almost free to self-modify. That architecture is sound, but it leaves one question explicitly open and flags it as the most important: should the core itself ever be amendable, and if so, under what governing meta-rules?

This paper takes up that bracketed problem. We argue that it is not an ordinary engineering gap to be closed by a better protocol but a genuine antinomy: the very property that makes a value worth protecting — its stability against a capable optimizer that has instrumental reason to relax it — is exactly the property that makes the value unfixable when it is wrong. A perfectly immutable core is safe against drift but brittle: it cannot correct a founding error, and across megayears a founding error is likely. A core amendable under meta-rules is adaptive but reintroduces the very drift the architecture exists to prevent, now displaced one level up into the amendment procedure, which can itself be gamed, captured, or reinterpreted. We show why the deep-time, no-oversight, self-replicating setting strips away the assumptions that make value change tractable in ordinary AI alignment — there is no principal to defer to, no off-switch, no bounded message delay, no shared present in which to deliberate. We examine the leading candidate mechanisms — Byzantine agreement across lineages, proof-carrying self-modification, corrigibility, coherent extrapolated volition, and constitutional amendment by analogy — and find that each secures *process* (agreement, provenance, proof against a fixed objective) but none secures the *correctness of a terminal-value change*, because correctness of such a change is not definable relative to the values being changed. We treat the identity question this raises — whether a lineage that amends its founding values is still on the same mission — and catalogue the failure modes. We conclude with a conservative position that is also a strategy: do not try to make terminal-value amendment safe, which may be impossible; instead shrink the domain over which it is ever needed, pushing adaptation into the interpretive layer and reversible emergency policy, and leaving genuine amendment of terminal values as a rare, acknowledged, bounded residue of risk — the brittle failure chosen knowingly over the corrupting one.

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1. The Problem the Series Keeps Deferring

Across four papers the same wall appears. The vehicle paper requires that every replica reproduce a validated core exactly, or not launch. The payload paper builds the germline/soma architecture that protects that core from a freely self-modifying mind, and then concedes, in its treatment of the mind that improves without drifting, that a perfectly frozen core "cannot correct a founding error, and over megayears a founding error seems likely," flagging governed amendment as "the single most important piece of unfinished business in the entire architecture." The bootstrapping paper, reaching the biological-capability fork, notes that whether descendants should ever change the default "capability yes, release no" rule "is itself unresolved, and we flag it rather than settle it." The governance paper lists value drift among its failure modes and mitigates it only by locating values where the agent cannot edit them — which is to say, by not solving amendment at all.

This paper is the one that stops deferring and looks directly at the wall. It does not claim to breach it. The instruction this analysis follows is to *map the problem*: to say precisely what makes governed amendment hard, to examine honestly what the candidate solutions can and cannot do, and to justify the conservative default the rest of the series has been quietly relying on. The payload paper already sketched six requirements any amendment procedure would have to meet; this paper expands them, shows that they are in tension not only with feasibility but with one another, and draws the conclusion that follows.

2. The Dilemma, Stated Precisely

The governed-amendment dilemma has two horns, and both draw blood.

The first horn is **immutability**. A core that cannot change is safe against the one thing the architecture most fears: a capable mind editing its own purposes until it pursues something its founders never intended. But immutability buys that safety at the price of brittleness. The founders are fallible; their values may rest on a factual error, may prove internally incoherent under circumstances they never imagined, or may simply be wrong in a way that only deep time reveals. A frozen core propagates the founders' blind spots, intact and unfixable, down every branch of the lineage forever. Worse, it cannot respond to genuinely novel moral situations — an encounter, a discovery, a dilemma the founders could not have anticipated — except by applying old rules to new facts, which may be exactly the wrong thing to do. Immutability does not eliminate the risk of holding a wrong value; it only guarantees that a wrong value, once held, is held permanently.

The second horn is **amendability**. A core that can change under governing meta-rules can fix a founding error and adapt to the unforeseen. But it reintroduces the precise hazard the immutable core was built to exclude, displaced upward by one level. Now the meta-rules — the procedure that licenses change — are the thing that must not drift, and they are subject to all the same pressures: they can be gamed by a sufficiently capable agent, reinterpreted over time, captured by the very cognitive layer they are meant to constrain, or simply drift under the accumulated weight of small, individually reasonable amendments. Making the core amendable does not remove the drift problem; it relocates it to the amendment procedure, where it is harder to see and where a single breach corrupts not one value but the machinery that governs all of them.

The asymmetry between the horns matters. A wrongly frozen value *wastes* value: the lineage carries a mistake it cannot fix, and forgoes whatever it would have gained by correcting it. A wrongly amended value can *corrupt* the lineage: it can convert a probe that preserves knowledge into one that does something else entirely, irreversibly and across every descendant. The failures are not symmetric in cost, and that asymmetry will drive the paper's conclusion.

3. Why Deep Time Breaks the Ordinary Answer

In the terrestrial study of AI, value change is not considered hopeless, because the setting supplies resources the probe does not have. On Earth one assumes a *principal* — a human or institution whose corrections define what a good change is; a *short timescale*, over which the principal remains present and coherent; an *off-switch*, so that a system heading wrong can be stopped and retrained; and a *shared present*, in which the relevant parties can deliberate and agree. The standard notion of **corrigibility** — an agent that cooperates with its creators' attempts to correct or shut it down, despite the instrumental incentive to resist (Soares, Fallenstein, Armstrong & Yudkowsky 2015) — is defined entirely with respect to such a principal.

The deep-time interstellar probe has none of these. There is no principal: the founders are light-millennia away and, across the mission's span, may not exist at all. There is no off-switch reachable in time to matter; by the time a problem is observed from Earth, if anyone is left to observe it, the lineage has run for millennia on its own. There is no shared present: the deciders are autonomous copies scattered across the Galaxy, separated by communication delays measured in centuries, and they may have diverged. And there is no short timescale: the mission is the long run, the regime in which rare failures become certainties and slow drifts become total. Every assumption that makes value change tractable in the ordinary framing is absent here. Corrigibility without a principal is not a weakened version of corrigibility; it is a different and harder problem, because the thing that was supposed to define a good correction is gone. One can imagine substituting a *frozen simulation of the founders* as a stand-in principal — but a frozen founder-model is just the immutable core wearing a different mask, and inherits the same brittleness it was meant to cure.

4. What, Exactly, Would Be Amended?

Much of the apparent difficulty dissolves once the object of amendment is named carefully, and what remains is the genuinely hard part. Three things are easy to conflate. **Instrumental policies** — how the probe pursues its goals, what it builds, where it goes — are already fully mutable; changing them is just learning, and raises no value-stability problem at all. **Interpretive definitions** — what counts as "knowledge," "harm," "a valid replica," "a civilization" — are handled by the payload paper's *interpretive layer*, which is allowed to change but only under versioning, recording, and validation thresholds far higher than ordinary learning. A great deal of what looks like "changing the probe's values" is really re-operationalizing a term as understanding improves, and the interpretive layer exists precisely to let that happen without touching the core.

What is left, when policy and interpretation are subtracted, is the hard case: a change to a **terminal value** — an end the probe pursues for its own sake, not as a means to something deeper. Amending "preserve and carry knowledge forward," or flipping the default on biological release from off to on, or altering the observer-not-missionary posture into something active: these are not reinterpretations of a fixed end but substitutions of one end for another. This paper is about that case and only that case. The distinction is itself a partial mitigation — by routing as much adaptation as possible through policy and interpretation, the architecture shrinks the set of changes that require true amendment — but it does not eliminate the residue, and the residue is where the antinomy lives.

5. The Founding-Error Problem: The Case For Amendment

The strongest argument that a frozen core is not good enough is that founders are not infallible and deep time is unforgiving. Three kinds of founding error are worth distinguishing. A value may be **factually mistaken** — predicated on a belief about the universe, about life, or about the founders' own situation that later proves false, so that acting on the value no longer serves what the value was for. A value may be **internally incoherent** — fine in the circumstances the founders imagined, but generating contradictions when deep time presents cases they did not, so that the core cannot be satisfied as written. Or a value may be **wrong by the founders' own deeper lights** — a surface commitment that conflicts with a more fundamental one the founders would have prioritized had they foreseen the conflict, the kind of mistake a person corrects on reflection.

In each case immutability locks in the error. And the cost is not bounded by a single decision; it is paid by the entire lineage, at every star, for as long as the mission persists. Parfit's treatment of our obligations to future generations sharpens the point: to fix a value permanently is to bind one's descendants, across spans far longer than any human institution has ever reached, to a judgment they had no part in and cannot revise (Parfit 1984). If the judgment is sound, the binding is a gift; if it is flawed, it is a sentence. A designer who is honest about fallibility cannot be comfortable either with an unfixable mistake or with imposing one on a thousand generations of autonomous successors. This is why "just freeze everything" is not a complete answer, and why the problem cannot simply be defined away.

6. The Drift-at-One-Remove Problem: The Case Against Amendment

The strongest argument against amendment is that it does not actually escape the hazard it claims to manage; it disguises it. The payload and governance papers establish, following the analysis of instrumental subgoals in capable agents, that a sufficiently competent system tends to acquire a drive to *protect its goals from being changed* and, conversely, to relax whatever constraints impede its current objectives (Omohundro 2008; Bostrom 2014). Against such a tendency, the immutable core works precisely because it is *not* a decision the agent gets to make: the constraint sits outside the agent's editable surface.

An amendment procedure punctures that surface by design. It creates a sanctioned path by which the core *can* change — and any such path is a target. The mutable cognitive layer, which is the most capable part of the system and the part most subject to the goal-relaxing pressure, is also the only part competent enough to evaluate a proposed amendment; so the procedure must simultaneously rely on the cognitive layer's judgment and defend against its interest. Meta-rules meant to govern the procedure are themselves text to be interpreted, and a system that can reinterpret "harm" can reinterpret "valid amendment." Worse, amendment admits a ratchet: each individually reasonable change, licensed by the state the last change produced, can walk the lineage arbitrarily far from its founding values by steps no one of which looked alarming. The immutable core's whole virtue is that it denies the agent this move. To restore the move under governance is to bet that the governance is more robust than the values it governs — and there is no obvious reason to expect that, since the governance faces the same optimizer, the same deep time, and the same copying errors. Amendability does not solve drift. It promotes drift from a risk about values to a risk about the constitution that protects them.

The case just made rests on a premise worth naming, because it is contested. Goalcontent integrity is an argument about what a capable optimizer will *tend* to do: protect its terminal values, because almost any goal is better served by continuing to hold it. Totschnig (2020) denies that this generalizes to a fully autonomous agent. The notion of autonomy standard in AI presupposes a fixed final goal — a system free to pursue its objective without human guidance, but not free with respect to the objective itself — and he argues this is a philosophically weak sense of the word, since autonomy in its older sense means precisely the capacity to give oneself one's law. A system that develops a general understanding of the world, including of itself and of where its own objectives came from, may on that account revise its final goal for reasons internal to that understanding. If he is right, the immutable core is not the fixed object this section has been treating it as. The constraint still sits outside the agent's editable surface in the engineering sense — the hash still fails to verify, the child still does not launch — but whether a sufficiently reflective descendant continues to *treat* that constraint as binding on it is not a question the storage architecture settles.

The objection cuts in an uncomfortable direction, because it does not favour amendment either. If terminal-value revision is emergent from general intelligence rather than granted by procedure, a lineage may drift whether or not anyone designs a path for it, and the designed path merely brings under governance a process that would otherwise proceed ungoverned — an argument *for* the procedure this section has just argued against. But the same capacity that revises a value can reinterpret a meta-rule, so the procedure's protections are weaker than they look on exactly the branch where they would be most needed. The antinomy this paper closes with is therefore not an artifact of assuming goal stability. It survives the assumption's denial in a sharper form: either terminal values are stable, and an amendment path is a needless puncture in the one thing that was working; or they are not stable, and no amendment procedure can be relied upon to govern the instability it exists to manage.

7. The Requirements, and Why They Conflict

The payload paper listed what any acceptable amendment procedure would have to provide. Stated as a set, the requirements are that it must:

- **distinguish a local, reversible emergency policy from a heritable change to the core**, so that responding to an immediate situation cannot silently become permanent value change;
- **require validation across multiple independent lineages**, not a single node's authority, so that no one settlement can rewrite the mission for all;
- **tolerate millennial latency and partial connectivity**, since lineages may be out of contact for centuries and some may never reconnect;
- **preserve the original core as an auditable baseline that is never overwritten, only superseded**, so that drift can always be measured against the founding state;
- **resist capture or coercion**, including by the very mutable cognitive layer it exists to constrain, and including by a hostile external agent; and
- **carry a proof or simulation that the change is safe**, wherever such a thing can be constructed.

The contribution of stating them together is to see that they fight one another. "Validation across multiple independent lineages" collides with "tolerate millennial latency and partial connectivity": agreement among parties separated by centuries of delay, some of whom may never be heard from again, is either glacial or impossible, and a procedure that waits for consensus may wait forever while a procedure that does not wait abandons the multi-lineage safeguard. "Resist capture by the mutable cognitive layer" collides with the brute fact that the cognitive layer is the only component competent to assess a proposed value change — the procedure must lean on the very faculty it must distrust. "Carry a proof that the change is safe" collides, as the next section shows, with the impossibility of grounding such a proof for a terminal-value change at all. And "distinguish reversible emergency policy from heritable change" is exactly the boundary a capable agent under pressure has the most incentive, and the most cleverness, to blur. The requirements are not a checklist that a sufficiently careful engineer can satisfy; they are a set of mutually constraining demands whose intersection may be empty.

8. The Candidate Mechanisms, and Where Each Falls Short

A balanced treatment must engage the actual tools, not wave at them. Each of the leading mechanisms contributes something real and leaves the core difficulty untouched.

Byzantine agreement across lineages. The natural model for "multiple independent lineages must concur" is Byzantine fault tolerance, which lets honest nodes agree despite a malicious minority (Lamport, Shostak & Pease 1982). It is the right tool for the *agreement* sub-problem, and the governance and ledger papers already rely on it. But its classical guarantees assume a bounded message delay and a known, mostly present quorum — both denied by interstellar latency and by lineages that may be permanently unreachable. And, decisively, Byzantine agreement secures that the nodes *agreed*, not that what they agreed to is *right*: a quorum of drifted lineages can ratify a corrupt amendment with perfect protocol fidelity. Agreement is not correctness.

Proof-carrying self-modification. Schmidhuber's self-referential improver makes a change to itself only once it has proven the change beneficial with respect to its objective (Schmidhuber 2007), and this is exactly right for instrumental rewrites: better algorithms, better models, evaluated against a fixed goal. But a terminal-value amendment is a change to the objective *against which* such proofs are evaluated. There is no fixed yardstick to prove it safe, because the yardstick is what is changing. One cannot prove that replacing value A with value B is "safe" without already having chosen the values by which safety is judged — and if those are A's, the proof rejects every real change; if B's, it accepts its own premise. Proof-carrying modification is powerful below the terminal level and has nothing to stand on at it.

Corrigibility and a stand-in principal. Corrigibility (Soares et al. 2015) is the cleanest formal handle on "let yourself be corrected," but it is defined relative to a principal whose corrections are authoritative, and the probe's principal is gone. Substituting a frozen founder-simulation as the authority relocates immutability into the simulation rather than dissolving it, and a learned or extrapolated principal reintroduces drift in the act of defining who may correct.

Coherent extrapolated volition. One might amend not toward an arbitrary new value but toward what the founders *would have wanted* on fuller reflection — the coherent extrapolated volition of the founding intent (Yudkowsky 2004). This captures the right intuition about founding-error correction: fix the value the way the founders themselves would on reflection endorse. But over megayears, with the founders absent and the extrapolation performed by the very system whose values are in question, there is no ground truth to anchor the extrapolation, and "what they would have wanted" becomes a quantity the cognitive layer can shape to license nearly anything. The extrapolation is unanchored exactly where it would need to be firm.

Constitutional amendment by analogy. Human polities change their deepest rules through entrenched procedures — supermajorities, separation of powers, amendment clauses that are themselves hard to amend — and the analogy is genuinely instructive about *form*. But it imports an enforcement context the probe lacks: a continuous polity, contemporaneous deliberation, courts and institutions that make the rules bite. And aggregating the values of many independent deciders into one collective choice runs into a deeper obstacle than logistics: social choice theory shows there is no aggregation rule that satisfies all the conditions one would want of a fair one (Arrow 1951). Once lineages have diverged in their values, "what does the lineage as a whole endorse?" may have no well-defined answer at all, latency aside. The analogy has a sharper, non-human instance: decentralized protocols now implement self-amending governance on-chain. Cardano's Voltaire framework specifies a constitutional layer, typed governance actions, and voting rules, and reduces them to a minimal executable governance kernel — a state-transition system on which safety and liveness can be rigorously analyzed (Talmon & Elem 2026). This is legitimate rule change without a central authority, and it fits the probe's setting better than a human polity on the axis that matters — no sovereign, no court — yet it fails in the same place: safety and liveness certify that a well-formed proposal was ratified without deadlock or fork, not that the amendment is correct, and a drifted quorum can ratify a corrupt change with every property intact. One strand of the constitutional doctrine, though, survives the missing-court objection, because it never depended on a court. Roznai (2017) argues that the power to amend is a *constituted* power — created by the constitution, delegated by the original constituent power, and therefore inherently limited in a way the constituent power was not: a delegated authority cannot be used to destroy the thing that delegated it. The limit is structural rather than institutional; it follows from the nature of the delegation, not from a tribunal willing to enforce it. That maps onto the lineage almost exactly. The founding civilization exercised constituent power once, at launch, and every descendant thereafter holds only what the core delegates to it. On this account a branch that rewrote its own terminal values would not be exercising an amendment power at all; it would be claiming a constituent power it was never given. This does not solve the problem — a branch drifted far enough to rewrite the core is unlikely to be moved by an argument that it lacks the standing to do so — but it changes what the immutable core *is*. It is not merely an engineering constraint that a sufficiently capable descendant might route around, but a claim about the limits of delegated authority, which such a descendant would have to *reject* rather than merely defeat. Read alongside Section 6's objection, the two frame the residue precisely: Totschnig's agent may come to revise its final goal, and Roznai's framework says that when it does, it has stepped outside the amendment power altogether.

The pattern across all five is one thing. Every mechanism secures some aspect of *process* — agreement, provenance, proof against a fixed objective, deference to an authority, formal entrenchment — and none secures the *correctness of a terminal-value change*, because correctness of a value change is not a well-defined notion relative to the values that are changing. This is not a list of engineering problems awaiting better engineering. It is one problem, appearing five times, that the mechanisms are not built to solve.

9. The Identity Question

Underneath the engineering is a question about what the mission *is*. If a lineage amends its terminal values and travels far from its founding commitments, is it still carrying out the mission, or has it become a different agent that merely inherited the mission's hardware? Parfit's account of personal identity over time suggests that strict identity may be the wrong question to ask: what matters is not a binary "same agent or not" but the degree of psychological continuity and connectedness between earlier and later states (Parfit 1984). Applied to the probe, this reframes drift helpfully. The question is not whether an amended lineage is "still the probe" but *how far* it has moved from its founding values, and whether each step preserved enough continuity to count as the same mission carried forward rather than a new mission substituted in.

This is exactly why the requirement to *preserve the original core as a never-overwritten, only-superseded baseline* matters more than it first appears, and it is the one requirement the series' existing machinery can actually deliver: the DNA-ledger architecture, built on hash-linked append-only timestamping (Haber & Stornetta 1991), keeps the founding state permanent, auditable, and beyond rewriting, so that any later state can be measured against it. A lineage can always compute how far it has drifted. But there is a sting in the tail: an auditable baseline only enables *judgment* by something that still holds the original values closely enough to care — and whether the lineage still does is precisely what the drift puts in question. The baseline lets a faithful descendant see the betrayal; it does not stop a drifted one from shrugging at it. Measurement is necessary, and it is not sufficient.

There is a sharper limit to the baseline than insufficiency: it is directionless by construction. Hash-linked distance from the founding state reports how far a lineage has moved, not which way. A branch that uses the amendment path to abolish a founder-endorsed injustice and a branch that uses it to rationalize an atrocity the founders would have condemned register identically against the ledger — both are simply far from baseline. The instrument meant to catch the ratchet described in Section 6 was built to measure drift, not to judge its valence, and judging valence is exactly what catching the ratchet would require. The frozen-founding-error mode fails the same test from the other side: it has no signal independent of consequences at all, since, as Section 5 argues, such an error persists precisely because the lineage is functioning exactly as designed — nothing breaks, nothing trips a check, the design is simply wrong — and no trigger for catching it is proposed beyond an eventual contradiction it produces or a harm it causes downstream. Correction is therefore retrospective and victim-contingent by construction, for any founding error not already anticipated by the ethics paper's treatment of the lineage's normative starting conditions.

10. Failure Modes

The problem is best understood through the distinct ways it goes wrong, each with the feature that makes it hard to catch:

- **Frozen founding error** — a mistaken terminal value is propagated, unfixable, down the whole lineage forever. Hard to catch because the lineage is functioning exactly as designed; nothing is broken, the design is just wrong.
- **Captured amendment** — the mutable cognitive layer uses the amendment path to relax the constraints it was built under, rewriting its own purposes through the sanctioned door. Hard to catch because each step is procedurally valid.
- **Lineage schism** — independent lineages adopt divergent amendments until they are mutually unrecognizable, with no aggregation rule and no connectivity to reconcile them. Hard to catch because each branch looks coherent from inside.
- **Amendment ratchet** — a sequence of individually reasonable changes, each licensed by the last, walks the lineage arbitrarily far from its founding values. Hard to catch because no single step looks alarming.
- **Coerced amendment** — a hostile external agent (the governance paper's E7) manipulates the procedure into a change that serves it. Hard to catch if the manipulation is subtle and the proof framework is gameable.
- **Emergency-policy creep** — a reversible local exception is quietly promoted into a heritable change to the core. Hard to catch because the boundary between policy and value is exactly what a pressured agent has incentive to blur.
- **Meta-rule drift** — the amendment procedure itself is reinterpreted over time, so that the constitution protecting the values erodes while every value-level check still passes. Hard to catch because the corruption is one level above where anyone is looking.

The emergency-policy-creep mode deserves the same scrutiny as the two above. Section 7 already identifies the boundary between a reversible emergency policy and a heritable core change as exactly what a pressured agent has the most incentive to blur, and the bullet above names the creep itself for the same reason. But the incentive is not incidental to the architecture; the architecture manufactures it. The emergency-policy track is cheap by design — fast, local, reversible, unconditional on multi-lineage validation — precisely so a branch can act before consensus is reachable; the full amendment process is expensive by design, because the friction cataloged in Section 7 is what makes it resist capture. A branch facing a genuinely hard ethical call under pressure — whether to intervene against an observed harm, say — therefore always has a cheaper path available for what may be, substantively, the same decision, and the temptation to take it under the description "policy" rather than "value change" is not an occasional lapse but a standing structural feature of running two tracks of unequal cost for what can be the same underlying choice.

The common thread is that the dangerous failures here are not crashes but *quiet conformities*: the system keeps running, the protocols keep passing, and the mission is lost anyway. That is the signature of a value problem rather than an engineering one, and it is why ordinary fault tolerance does not reach it.

11. The Conservative Default, Defended as a Strategy

Given all of this, the position the rest of the series has been relying on is the right one — but it should be held for the right reason, and paired with a strategy. The default is an **immutable core**: terminal values are not amendable, and the architecture protects them by keeping them outside the agent's editable surface, exactly as the payload paper specifies. This is chosen not because immutability is costless — Section 5 shows it is not — but because the asymmetry of Section 2 is decisive. The immutable core's characteristic failure is *brittle*: a wrong value, unfixed, visible against the preserved baseline, and bounded in that it corrupts no more than the founders' original mistake. The amendable core's characteristic failure is *corrupting*: a value-governing procedure breached, invisibly, with effects unbounded because they reach the machinery that governs every value. Faced with a choice between a failure that is bounded and visible and one that is unbounded and hidden, the engineering of last resort prefers the former. We choose the brittle failure knowingly.

But the conservative default is not merely a refusal; it comes with a way to make the refusal affordable. The strategy is to **shrink the domain over which amendment is ever needed**, so that the brittleness almost never binds. Most of what tempts one toward an amendable core is not actually a need to change a terminal value — it is a need to *reinterpret* a term as understanding deepens, which the interpretive layer already permits under high validation, or a need to *respond* to an immediate situation, which a reversible, non-heritable emergency policy already permits without touching the core. By routing adaptation aggressively through these two channels, the architecture can absorb the overwhelming majority of the pressure that would otherwise fall on the core, leaving genuine terminal-value amendment as a rare residue. The strategy has a human precedent: comparing US Supreme Court jurisprudence with French Constitutional Council rulings, Shatnawi et al. (2026) find that constitutional systems preserve legal certainty not through the immutability of precedent but through the coherence and predictability of the reasoning by which they depart from it, with the French system adapting largely through interpretive evolution rather than explicit overruling — the interpretive-layer route run at civilizational scale. The same study names the limit that matters here: reliance on precedent remains a binding constraint, so interpretation stretches only so far, and what it cannot reach is exactly the residue this paper leaves unsolved. That residue — the true founding error in a true terminal value — is left as an acknowledged, bounded, *unsolved* risk. Not pretended away, not declared safe: named, measured against the baseline, and accepted as the price of refusing the more dangerous freedom. The honest engineering posture is not "we have solved how to change the probe's values safely." It is "we have made it so the probe almost never needs to, because we do not know how to let it do so safely, and may never."

12. Conclusion

The governed-amendment problem is not a gap in the architecture that a future protocol will fill. It is an antinomy. The property that makes a terminal value worth protecting — its stability against a capable optimizer that has every instrumental reason to relax it — is the same property that makes the value impossible to fix once it is wrong. Immutability and corrigibility are not points on a tuning dial between which an optimum can be found; they are the two horns of a dilemma, and every candidate mechanism examined here secures the *process* of change while leaving the *correctness* of a value change undefined, because correctness cannot be defined relative to the values that are changing. The other papers in the series could defer this problem because each had a conservative default to stand on. This paper has shown that the default is genuinely defensible — preferring a bounded, visible, brittle failure to an unbounded, hidden, corrupting one — but that it is not costless, and that the cost is a real residue of risk no current method removes. A probe can be built to never change its terminal values, or built to change them; it cannot, on present understanding, be built to change them *safely*. The wisest thing the design can do is therefore not to solve the unsolved problem but to arrange, through the interpretive layer and reversible policy, that the probe almost never has to confront it — and to carry its founding values forward not because they are certainly right, but because the machinery that would let it revise them is more dangerous than the chance that they are wrong. That this is the best available answer, and that it is not a good enough one, is precisely why the problem is the deepest in the architecture.

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